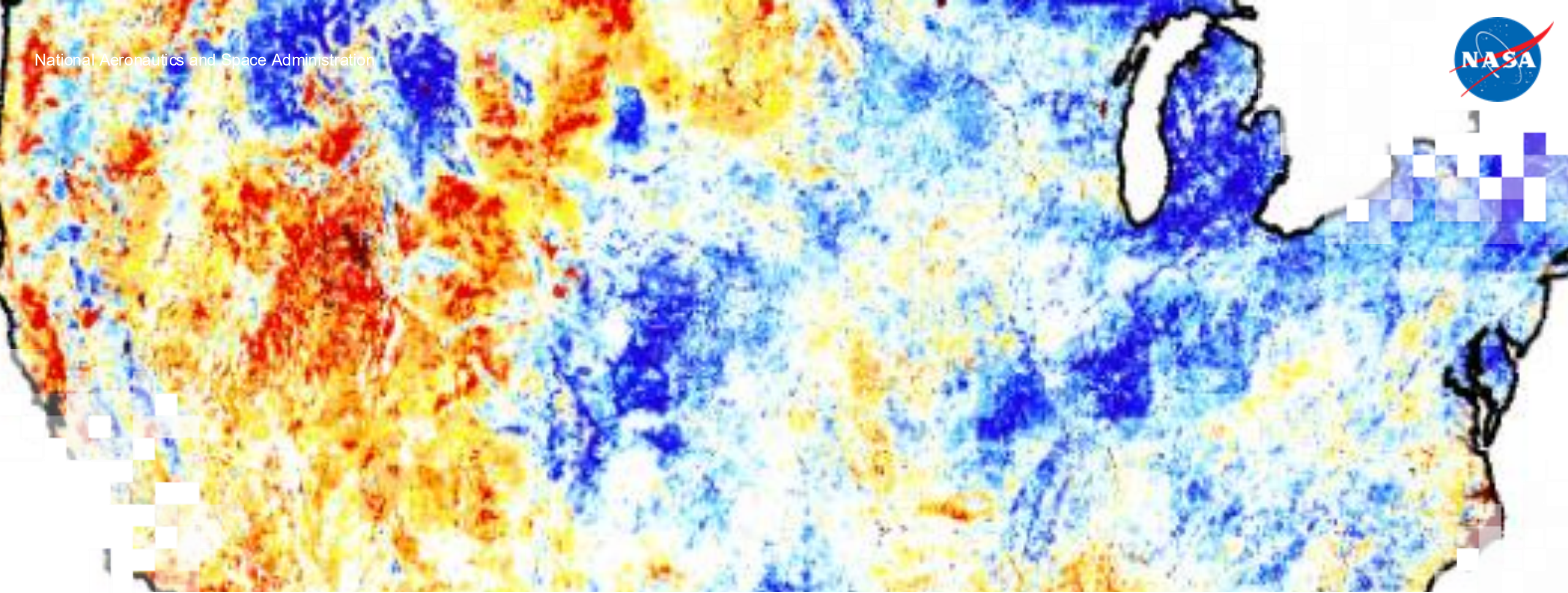


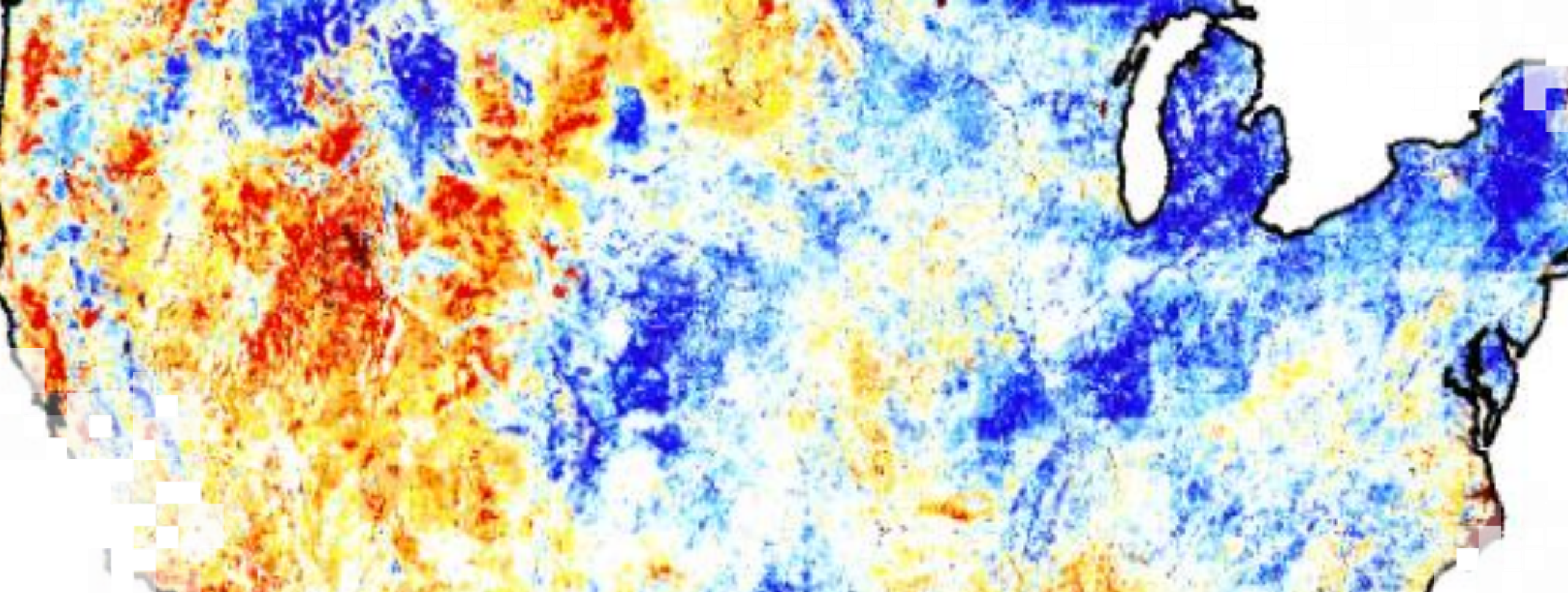


Agricultural Flash Drought Detection using Solar-Induced Fluorescence (SIF) and Soil Moisture Data

Session 1: Introduction to Flash Drought Monitoring using SIF and Soil Moisture

Nicholas Parazoo (JPL/Caltech), Jacqueline Ryan (JPL/.Caltech)

September 24, 2026



About ARSET

About ARSET

- **ARSET provides accessible, relevant, and cost-free training on remote sensing satellites, sensors, methods, and tools.**
- Trainings include a variety of applications of satellite data and are tailored to audiences with a variety of experience levels.



AGRICULTURE



DISASTERS



ECOLOGICAL CONSERVATION



HEALTH & AIR QUALITY



WATER RESOURCES



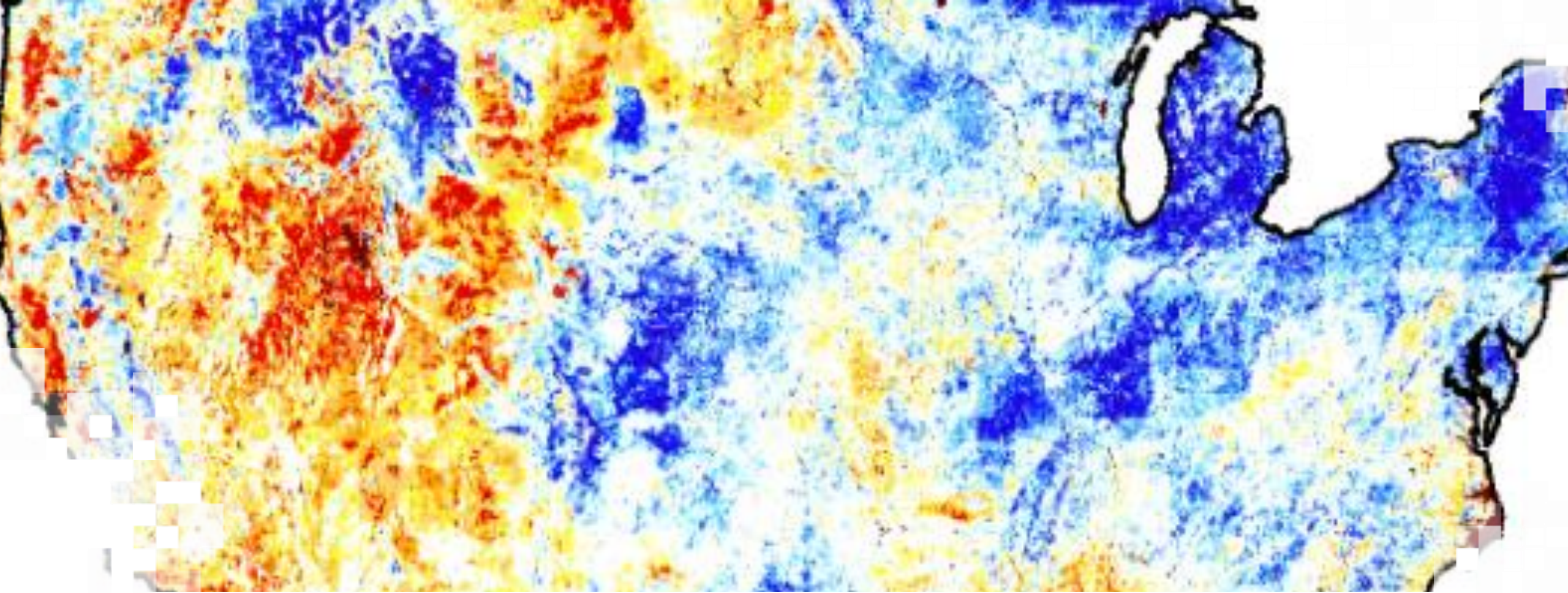
WILDLAND FIRES



About ARSET Trainings

- Online or in-person
- Live and instructor-led or asynchronous and self-paced
- Cost-free
- Bilingual and multilingual options
- Only use open-source software and data
- Accommodate differing levels of expertise
- Visit the [ARSET website](#) to learn more.





Agricultural Flash Drought Detection using Solar-Induced
Fluorescence (SIF) and Soil Moisture Data
Overview

Training Outline

Session 1
**Introduction to
Flash Drought
Monitoring using SIF
and Soil Moisture**

Sep. 24, 2026

Session 2
**Flash Drought
Detection and
Prediction**

Sep. 29, 2026

Homework

Opens Sep. 29 – Due Oct. 20, 2026 – Posted on Training Webpage

A certificate of completion will be awarded to those who attend all live parts and complete the homework assignment(s) before the given due date.

Prerequisites

- [Fundamentals of Remote Sensing](#)
- [Use of Solar Induced Fluorescence and LIDAR to Assess Vegetation Change and Vulnerability](#)
- [Solar Induced Fluorescence \(SIF\) Observations for Assessing Vegetation Changes Related to Floods, Drought, and Fire Impacts](#)

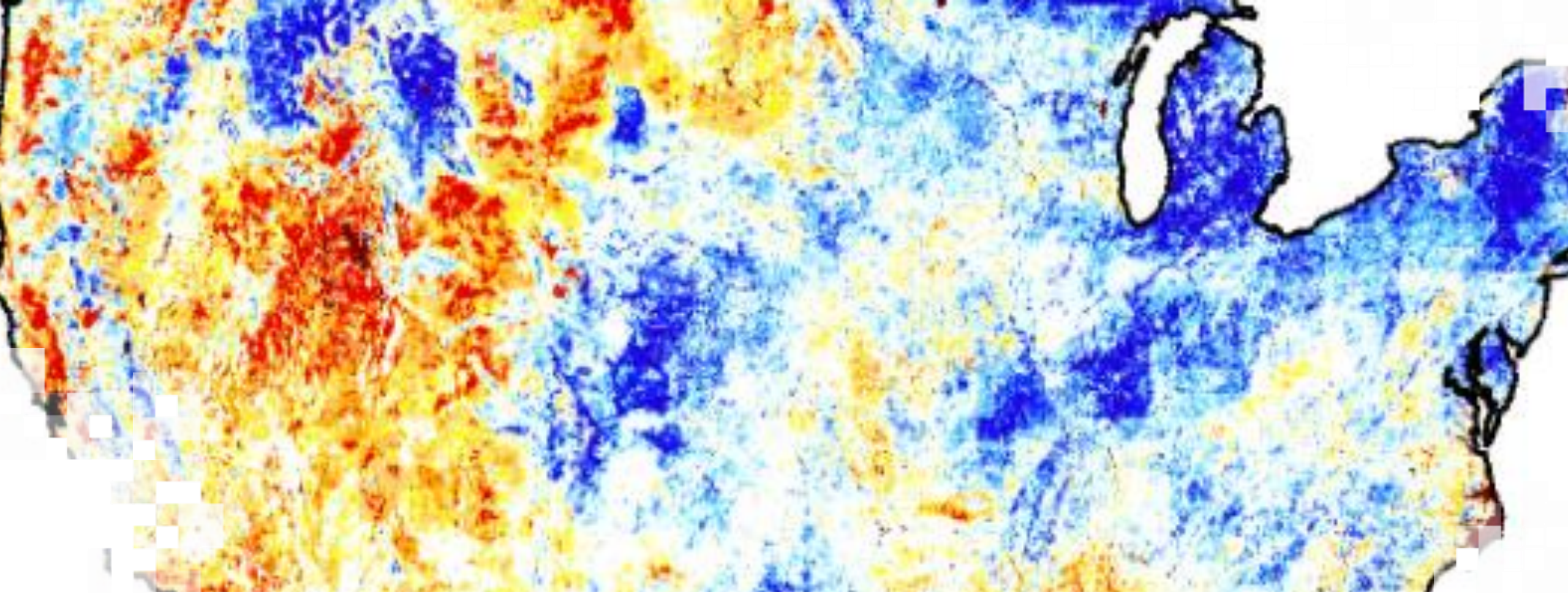


Training Learning Objectives

By the end of this training, participants will be able to:

- Identify the causes, risks, and consequences of flash droughts as compared to slowly- evolving droughts.
- Recognize how SIF data can be used for the detection and prediction of flash drought events.
- Compare the advantages and limitations of SIF, soil moisture, and meteorological data to detect the onset of a flash drought.
- Synthesize Solar-Induced Fluorescence Rapid Change Index (SIF-RCI) time series data for a selected region using a provided Jupyter Notebook.
- Retrieve and manipulate SMAP Soil Moisture (SM) and land process model-based Soil Water Deficit Index (SWDI) time series and compare with SIF-RCI time series data using a provided Jupyter Notebook.
- Compare the implications of different methods for flagging and identifying flash droughts in real-world scenarios.





Session 1:
Introduction to Flash Drought Monitoring using SIF and Soil
Moisture

Session 1 – Instructors

Erika Podest

Scientist and ARSET Instructor
NASA Jet Propulsion Laboratory,
California Institute of Technology



Dr. Nick Parazoo

Discipline Program Manager,
Carbon Cycle and Ecosystems
NASA Jet Propulsion Laboratory,
California Institute of Technology



Jackie Ryan

Data Visualization Developer
NASA Jet Propulsion Laboratory,
California Institute of Technology



Session 1 Objectives

By the end of Session 1, participants will be able to:

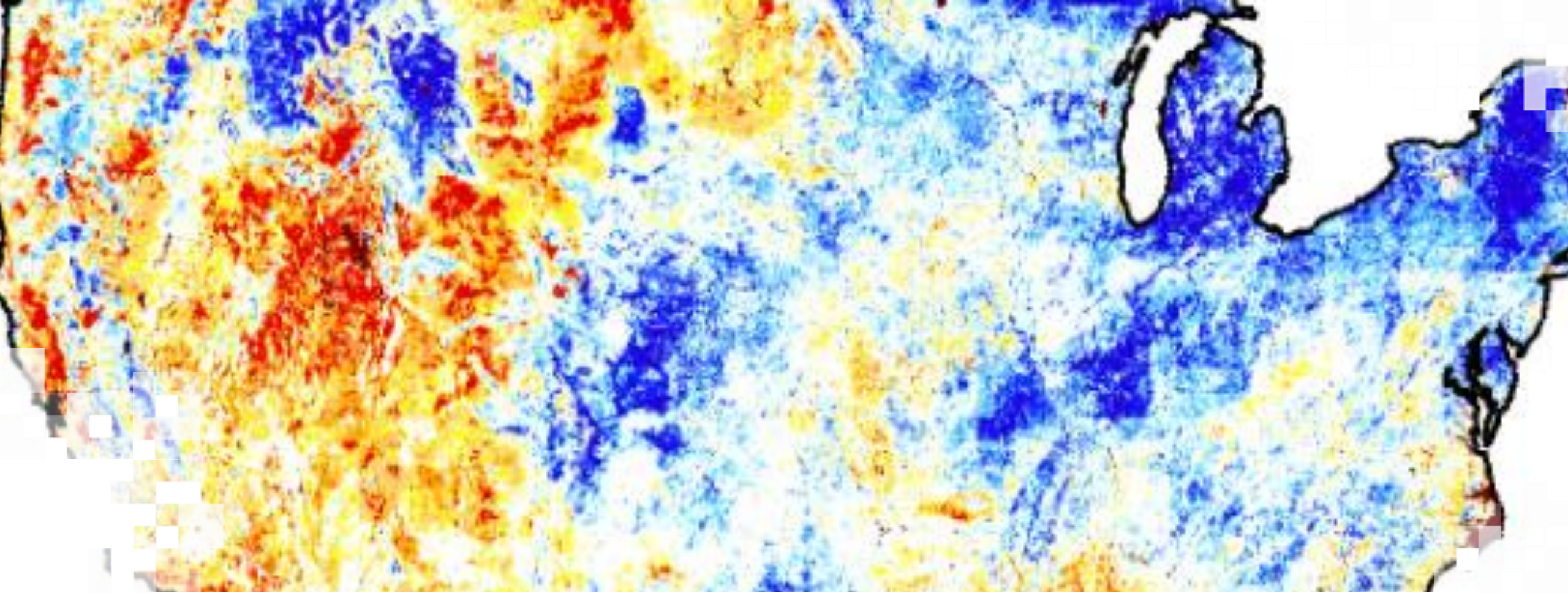
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- Synthesize Solar-Induced Fluorescence Rapid Change Index (SIF-RCI) time series data for a selected region using a provided Jupyter Notebook.



How to Ask Questions

- Please write your questions in the Questions box and we will answer them at the end of the webinar.
- Feel free to enter your questions as we go. We will try to respond all the questions during the Q&A session after the webinar.
- The remainder of the questions will be answered in the Q&A document, which will be posted to the training website about a week after the training.

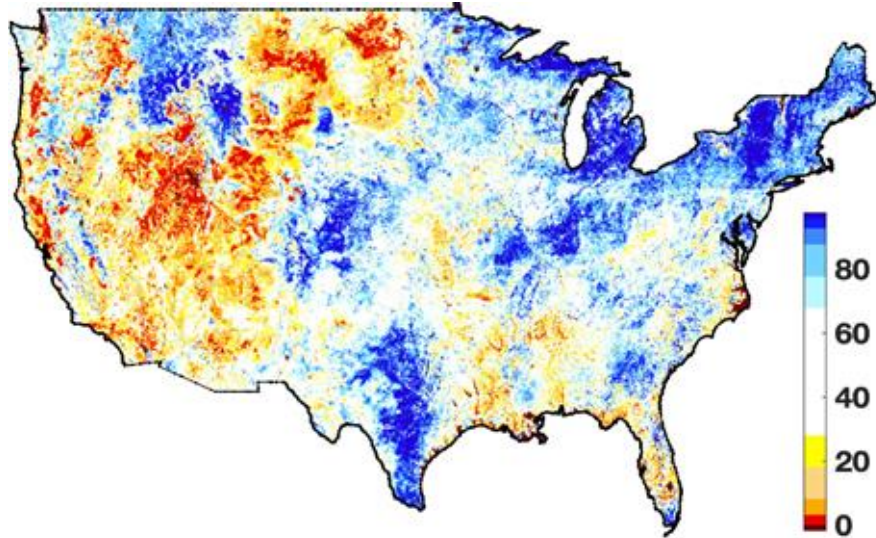




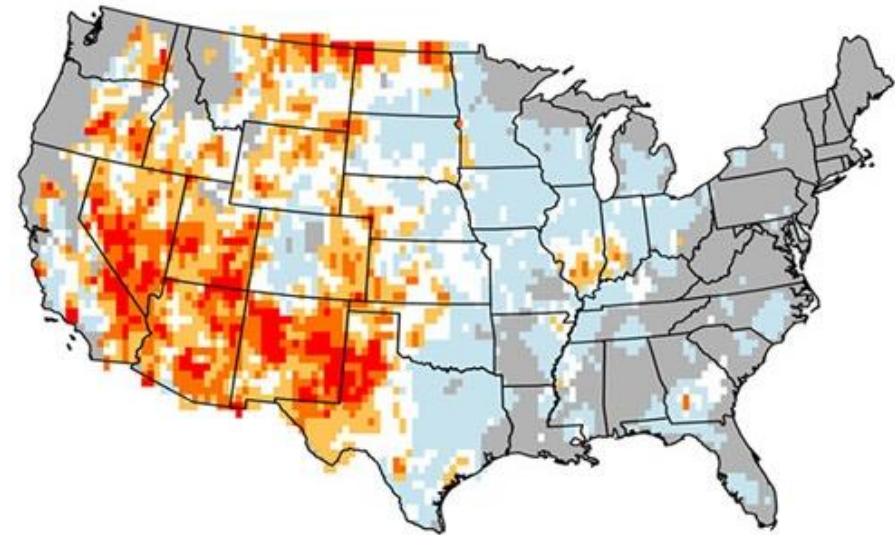
Session 1:
Introduction to Flash Drought Monitoring using SIF and Soil
Moisture

Part 1 Objectives

Orbiting Carbon Observatory (OCO-2) Solar Induced Chlorophyll Fluorescence (SIF)



Soil Moisture Active Passive (SMAP) Soil Moisture



OCO-2 and SMAP Flash Drought Indicators

- Review drought types and drought indices
- Identify how drought is typically detected
- Explore satellite remote sensing products as novel indicators of flash drought



Glossary

OCO-2: Orbiting Carbon Observatory

SMAP: Soil Moisture Active Passive

SIF: Solar-Induced Chlorophyll Fluorescence

SM: Soil Moisture

RZSM: Root Zone Soil Moisture

SPI: Standardized Precipitation Index

SPEI: Standardized Precipitation Evapotranspiration Index

PDSI: Palmer Drought Severity Index

USDM: United States Drought Monitor

FDSI: Forest Drought Stress Index

EDDI: Evaporative Demand Drought Index

Propagation Ratio: Severity of hydrological drought relative to meteorological drought

Ecosystem Services: Benefits that ecosystems provide to nature and humans

Z-Score: Standardized anomaly of a quantity (e.g., *SIF*)

RCI: Rapid Change Index (*used for detection of change, e.g., SIF or SM*)



Drought is defined into different types.

Meteorological Drought: Period of low precipitation

Hydrological Drought: Period of inadequate soil water resources to meet established use

Agricultural Drought: Period of declining soil moisture with impacts to crops

Ecological Drought: Episodic deficit in water availability impacting ecosystem function, services, and feedbacks

Snow Drought: Period of abnormally low snowpack

Socioeconomic Drought: Failure of water resources to meet water demands



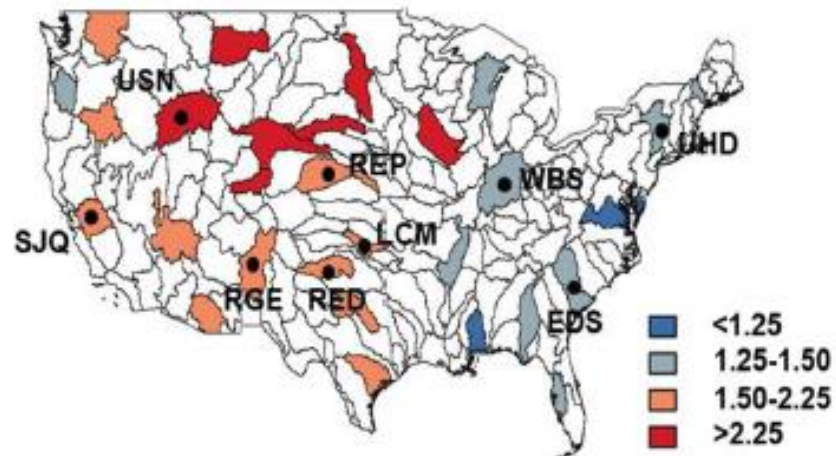
Wilhite & Glantz 1985; AMS 2004; Mishra & Singh 2010; Crausbay et al 2017; *drought.gov

NASA ARSET – Agricultural Flash Drought Detection using Solar-Induced Fluorescence (SIF) and Soil Moisture Data: Part 1

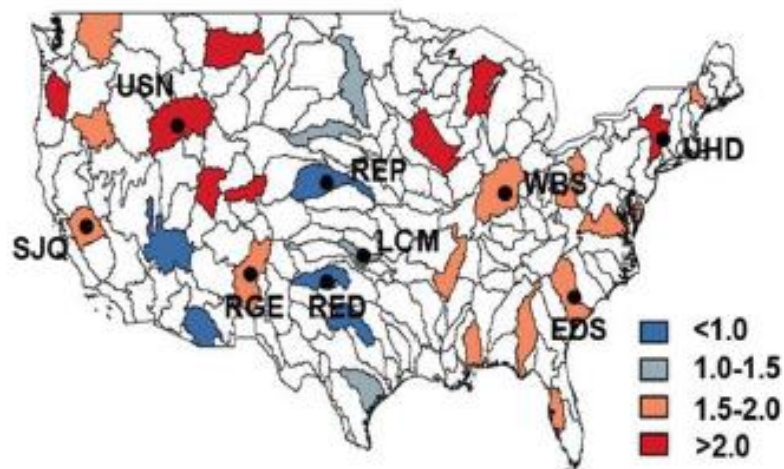


Connection between Different Drought Types

(a) Propagation ratio for soil moisture droughts



(b) Propagation ratio for hydrologic droughts



- Meteorological droughts have strong impact on soil moisture and hydrological droughts over the Western US.
- Subsequent impacts to ecological and agricultural drought.
- Maps and drought indices help communicate the intended drought information.

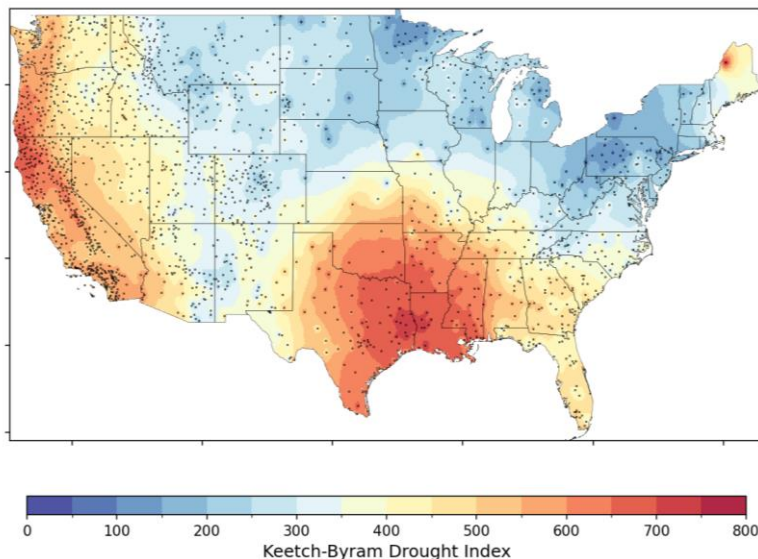
Apurv and Cai 2021. Source: Dr. Rong Fu, UCLA



Drought indicators provide a user-friendly visualization of impact.

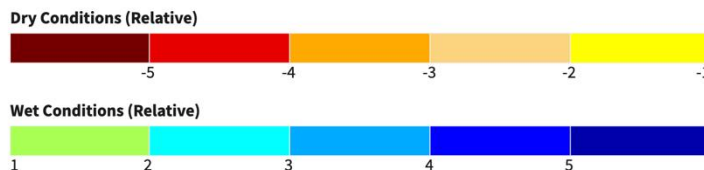
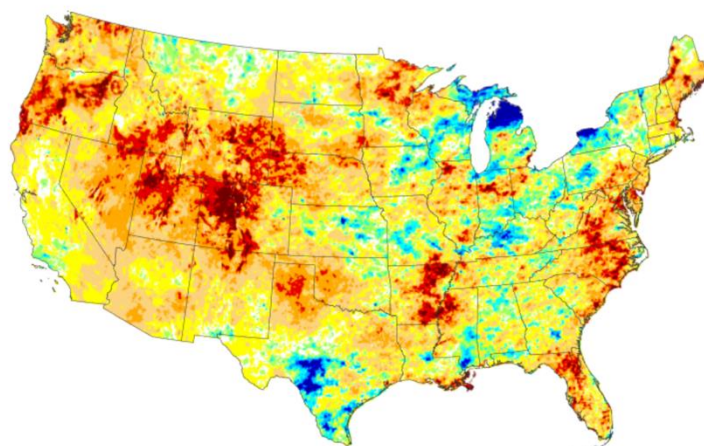
Keetch-Byram Drought Index (Wildfire Risk)

Keetch-Byram Drought Index (KBDI) (2026-08-24) (Observed)



Palmer Drought Severity Index (Agricultural and Water Supplies)

Palmer Drought Severity Index (PDSI)

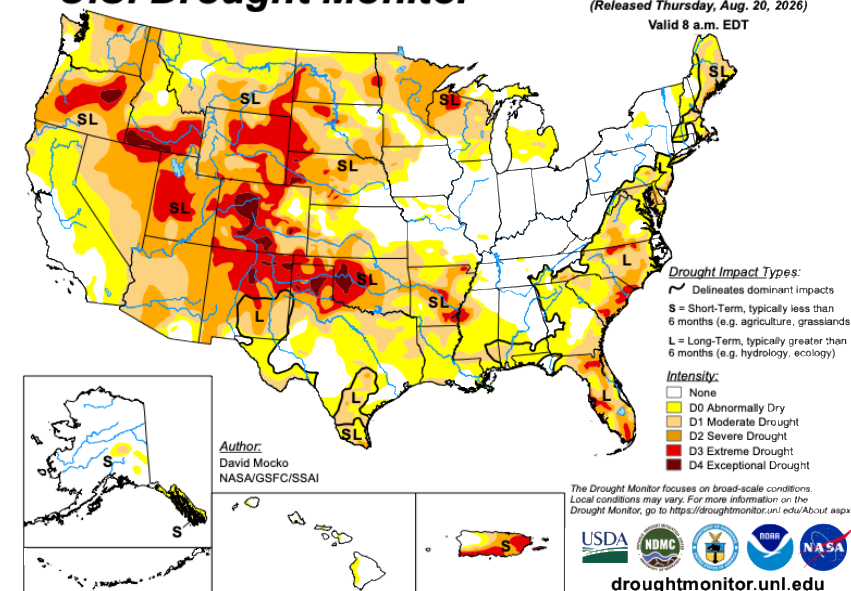


Drought.gov

U.S. Drought Monitor (Operational Decision Support)

U.S. Drought Monitor

August 18, 2026
(Released Thursday, Aug. 20, 2026)
Valid 8 a.m. EDT



Other Commonly Used Operational Drought Indices:

- Standardized Precipitation Index (SPI) -> [Best suited for wet climate](#)
- Standardized Precipitation Evapotranspiration Index (SPEI) -> [Needed for warmer and dry climate](#)



USDM incorporates multi-source information into simple, readable drought categories.

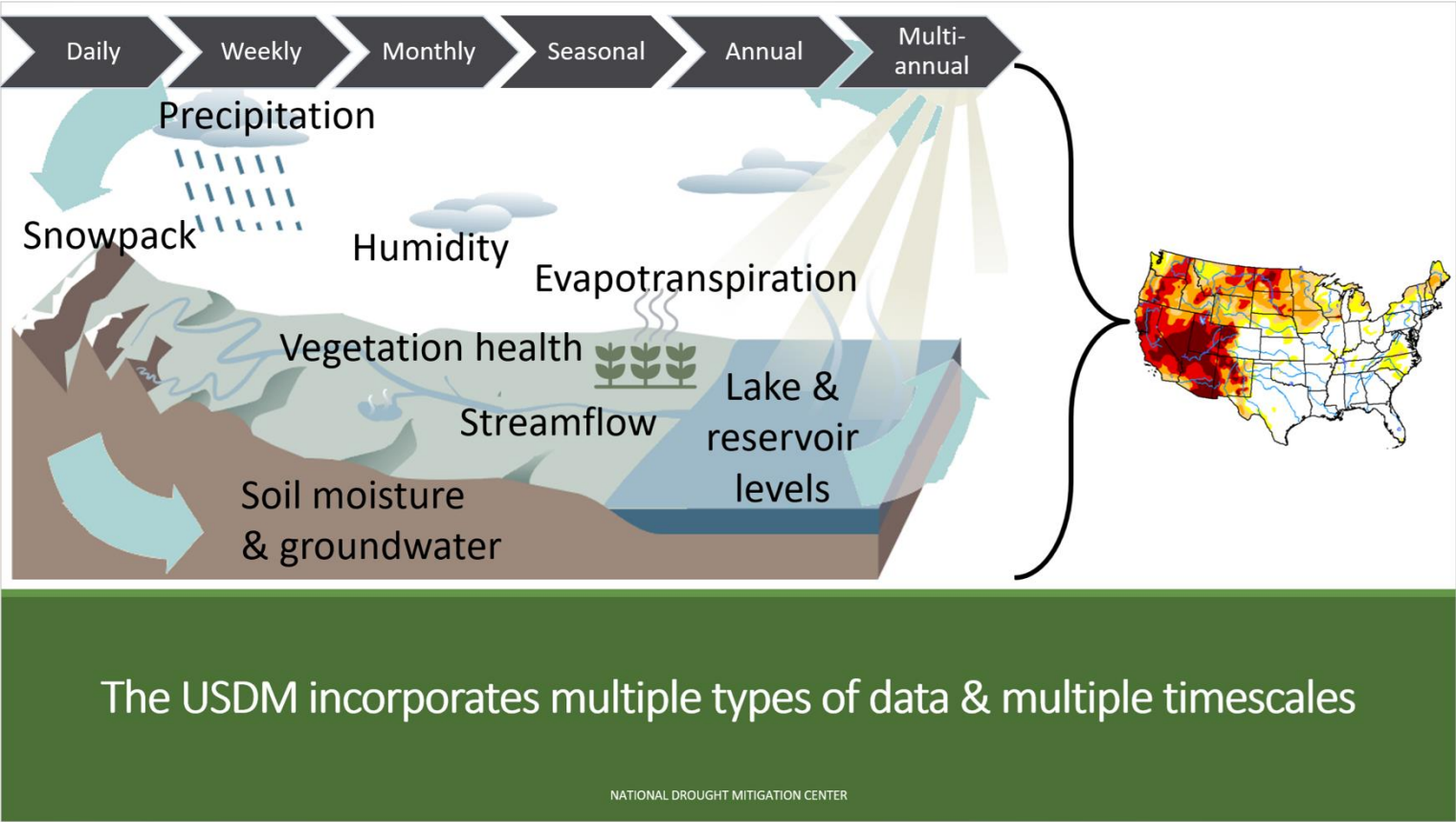
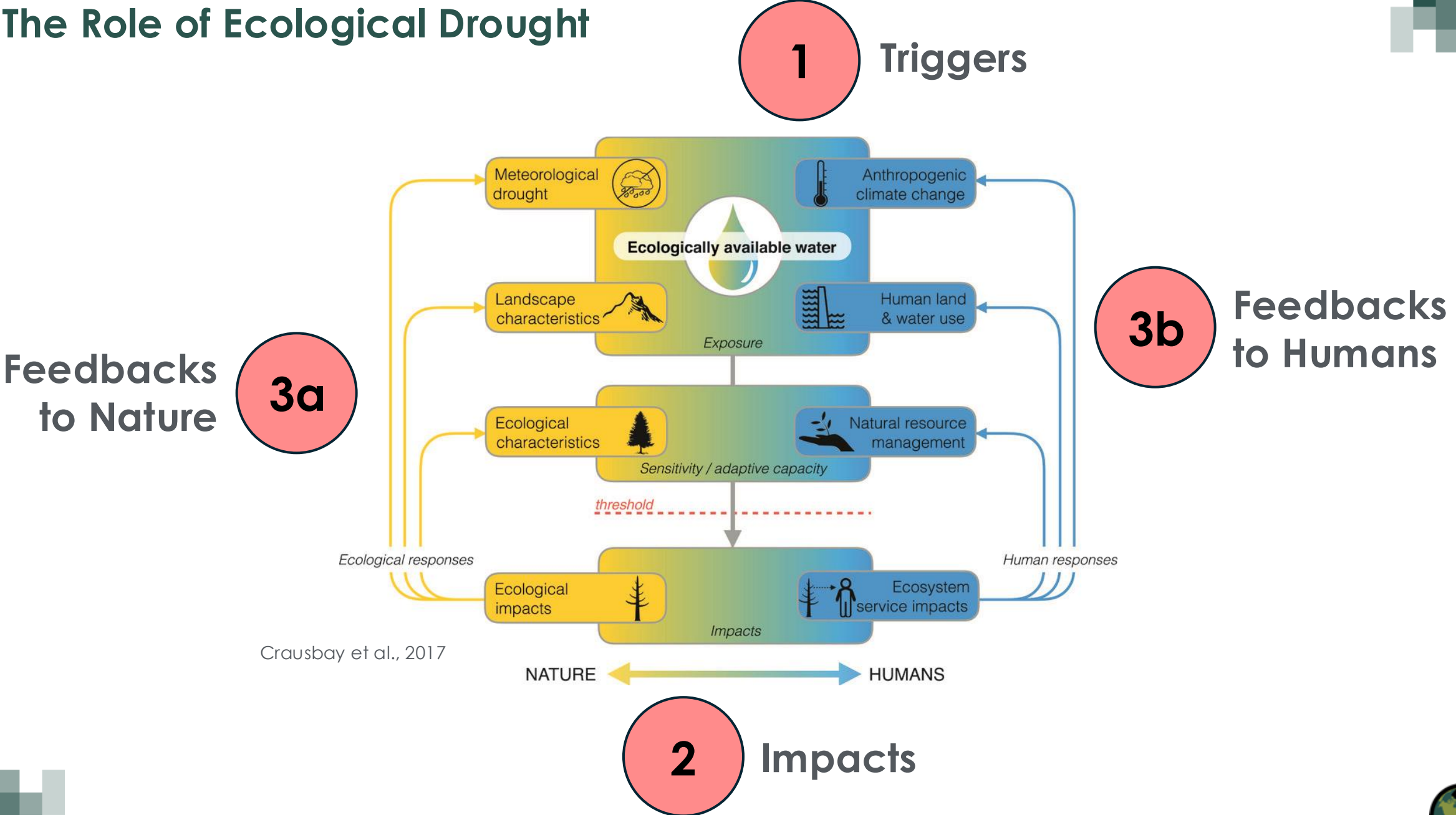


TABLE 2. The association of the six key objective drought indicators with the magnitude of drought severity in the Drought Monitor.

Drought Monitor classification							
Drought type		Associated ranges of objective indicators					
Category	Description	Palmer drought	CPC soil moisture	USGS weekly	Percent of normal	Standardized precipitation	Satellite vegetation
D0	Abnormally dry	-1.0 to -1.9	21-30	21-30	< 75% for 3 months	-0.5 to -0.7	36-45
D1	Moderate drought	-2.0 to -2.9	11-20	11-20	< 70% for 3 months	-0.8 to -1.2	26-35
D2	Severe drought	-3.0 to -3.9	6-10	6-10	< 65% for 6 months	-1.3 to -1.5	16-25
D3	Extreme drought	-4.0 to -4.9	3-5	3-5	< 60% for 6 months	-1.6 to -1.9	6-15
D4	Exceptional drought	-5.0 or less	0-2	0-2	< 65% for 12 months	-2.0 or less	1-5

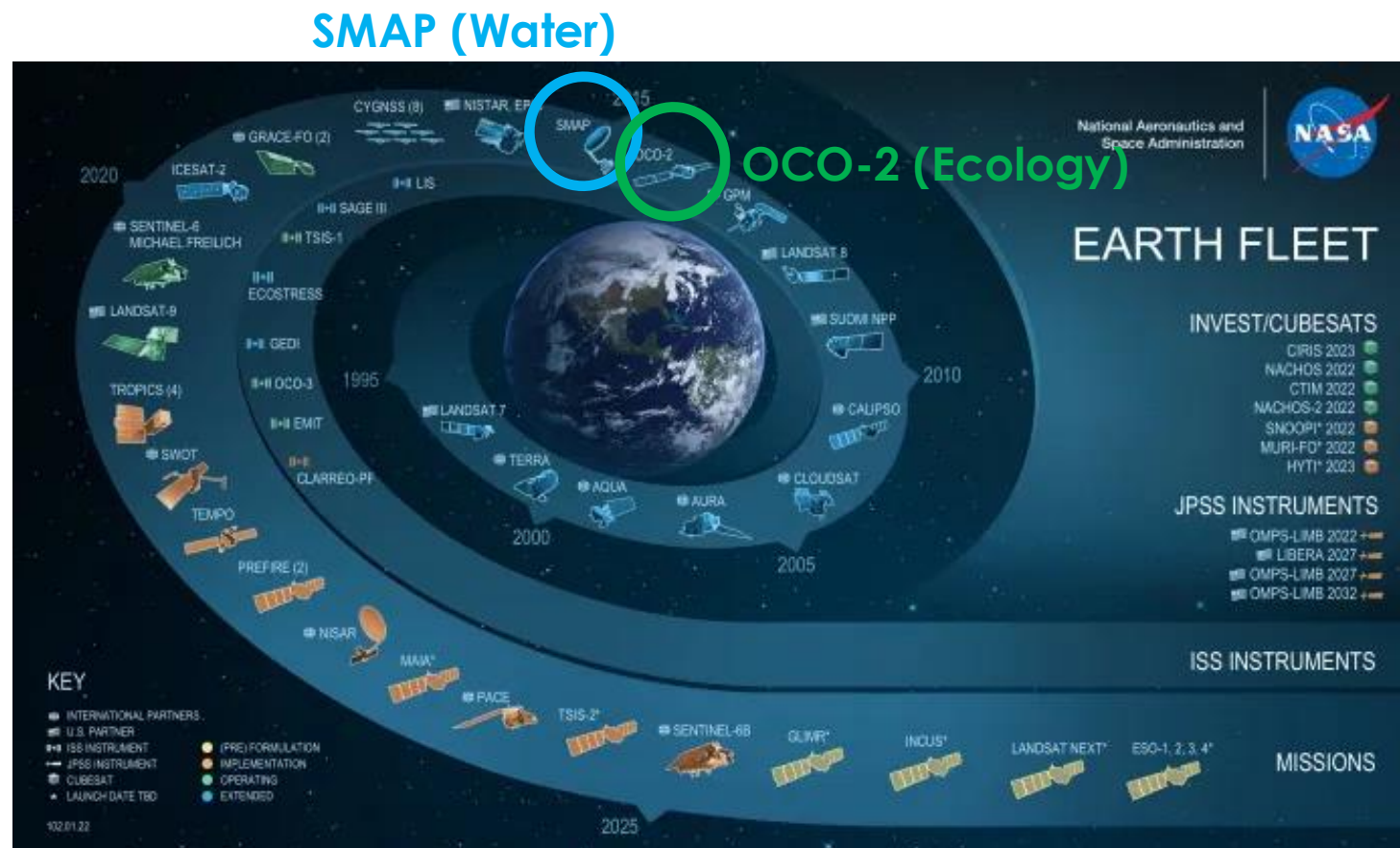
Svoboda et al., 2002

The Role of Ecological Drought



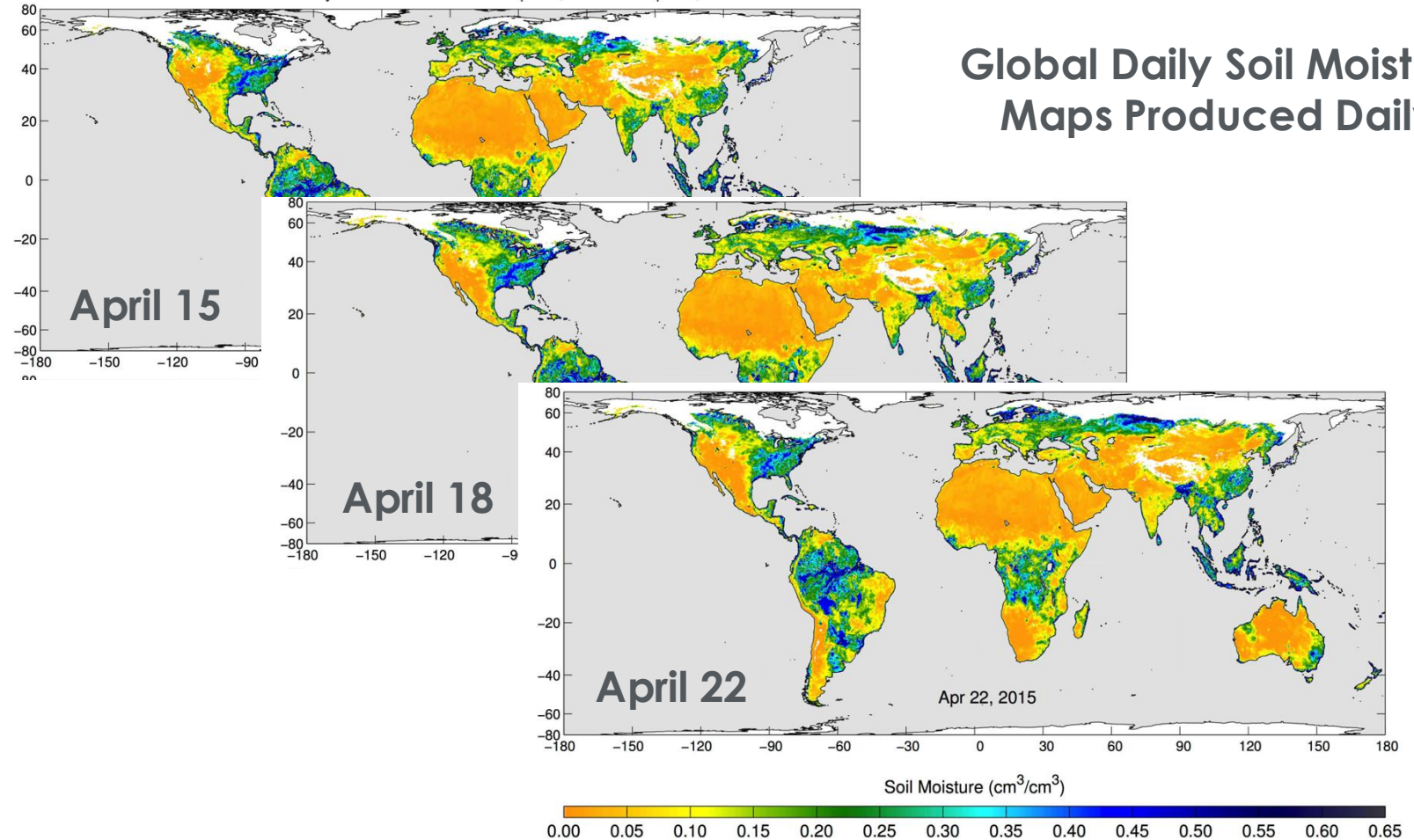
Drought indices are typically derived from satellite data.

- Earth observing satellites are diverse
- Daily to weekly monitoring of global land surface
- A growing number of satellites focus on water and ecology
- **OCO-2** and **SMAP** are ideally suited for flash drought detection
- Record is 12 years and counting



SMAP Captures View of Global Soil Moisture

- SMAP radiometer is sensitive to global near-surface soil moisture (~centimeters).
- Deeper soil moisture (~meters) can be estimated by integrating radiometer soil moisture data with models.
- **Important:** Dense vegetation obscures soil signal.



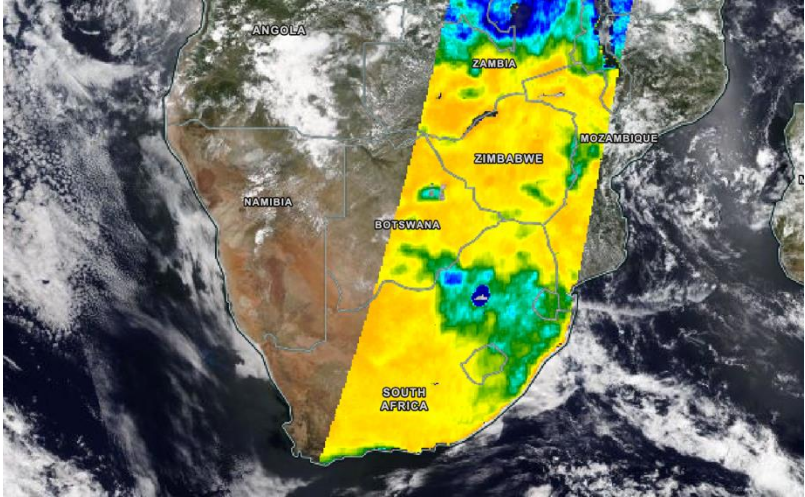
<https://smap.jpl.nasa.gov/observatory/instrument/>

<https://smap.jpl.nasa.gov/resources/87/smaps-radiometer-captures-views-of-global-soil-moisture/?category=images>

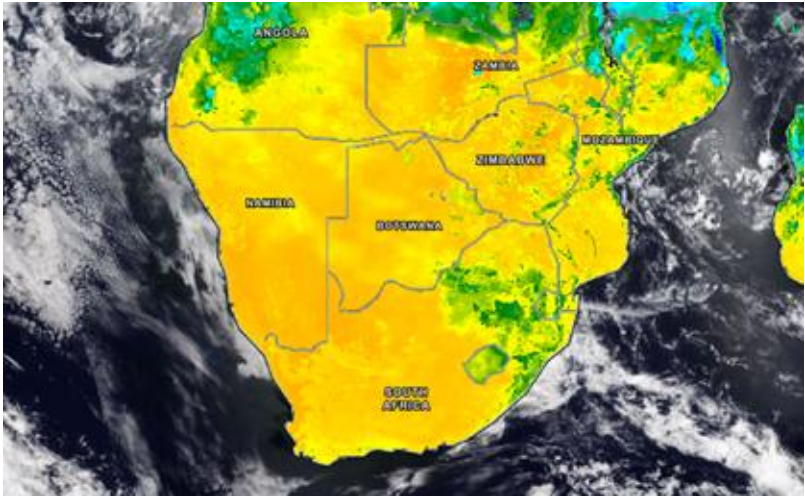
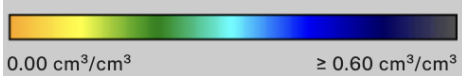


Modeled vs. Observed Soil Moisture

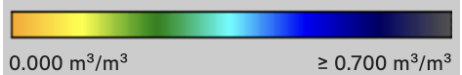
	Advantages	Disadvantages
Observations	• Lower uncertainty • Skill is closer to in-situ measurement than models	• Spatial & temporal gaps • Only available for surface layer
Models (Note: SMAP L4 data assimilates observations)	• Spatially contiguous • Even temporal spacing • Can estimate root zone	• Higher uncertainty • Biases in assumptions about soil



SMAP L3 Surface SM
(Observed)



SMAP L4 Surface SM
(Modeled)

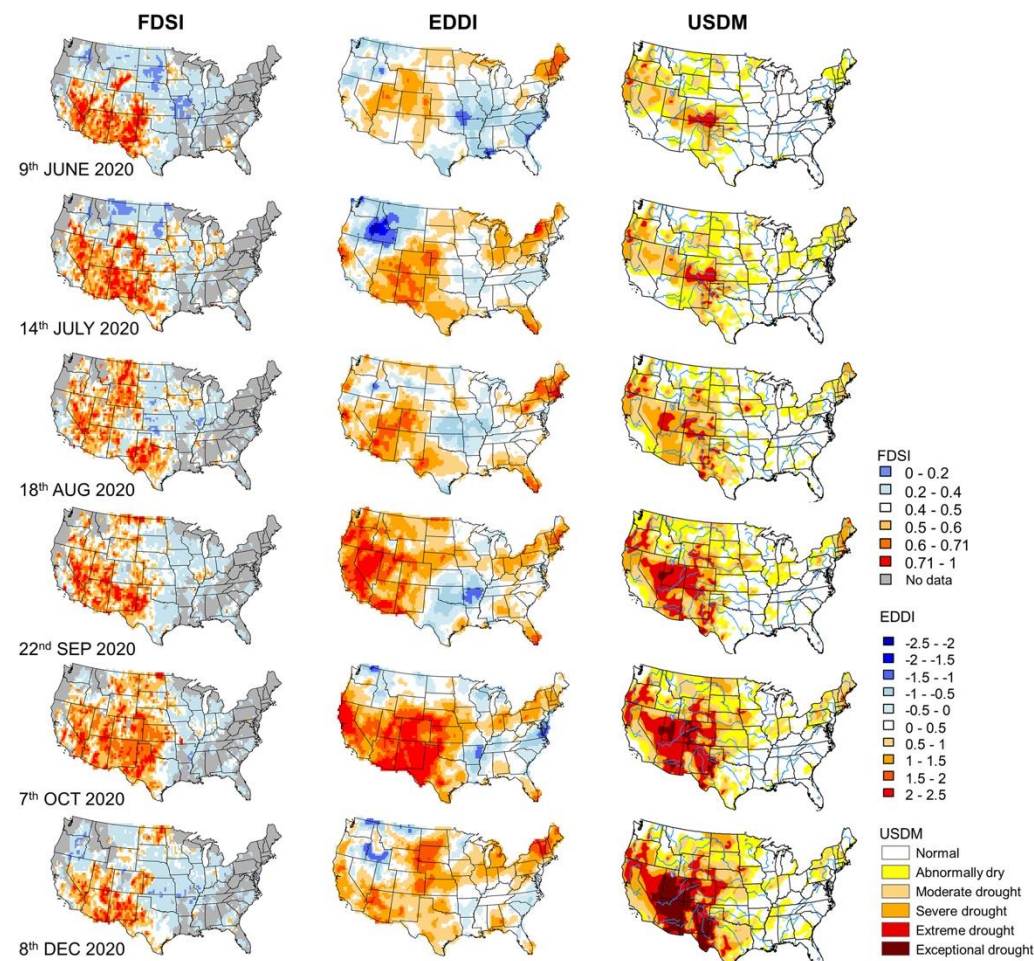
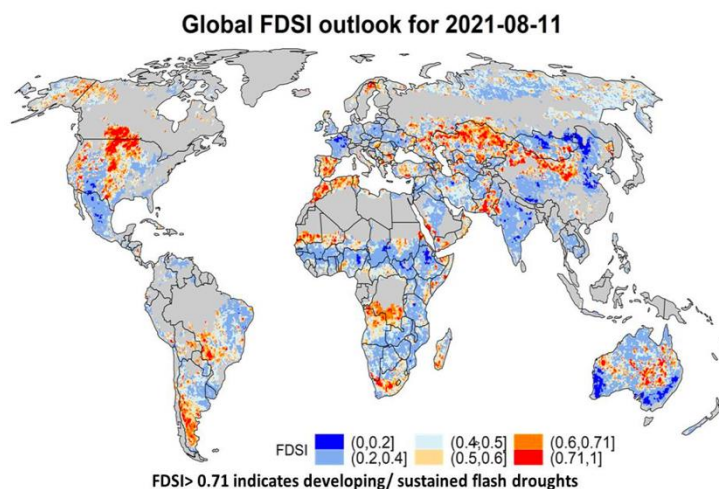


<https://go.nasa.gov/3UINFuW>



Global Flash Drought Monitoring using SMAP Soil Moisture

- **Flash Drought Stress Index (FDSI)**, based on SMAP L3 SM, tracks soil moisture state (stress) and rate of soil moisture drawdown (intensification)
- Corresponds with **Evaporative Demand Drought Index (EDDI)** and **U.S. Drought Monitor (USDM)**
- Global satellite data offers global, near-real time monitoring of flash drought



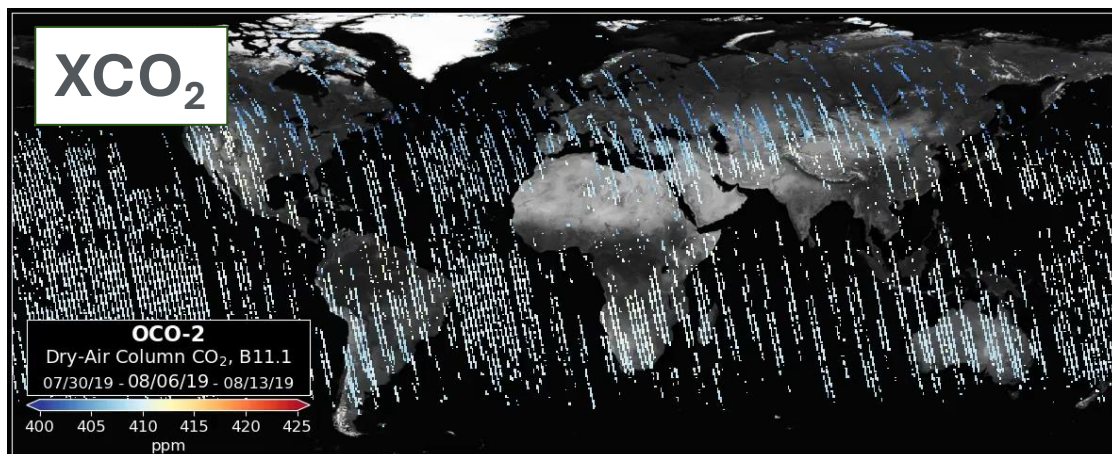
https://smap.jpl.nasa.gov/internal_resources/564/

Sehgal et al, 2021, WRR



OCO-2: Watching the Earth Breathe from Space

OCO-2 has been a pathfinder in understanding how carbon moves between reservoirs in the Earth system.



Column-Integrated Carbon Dioxide (XCO₂)

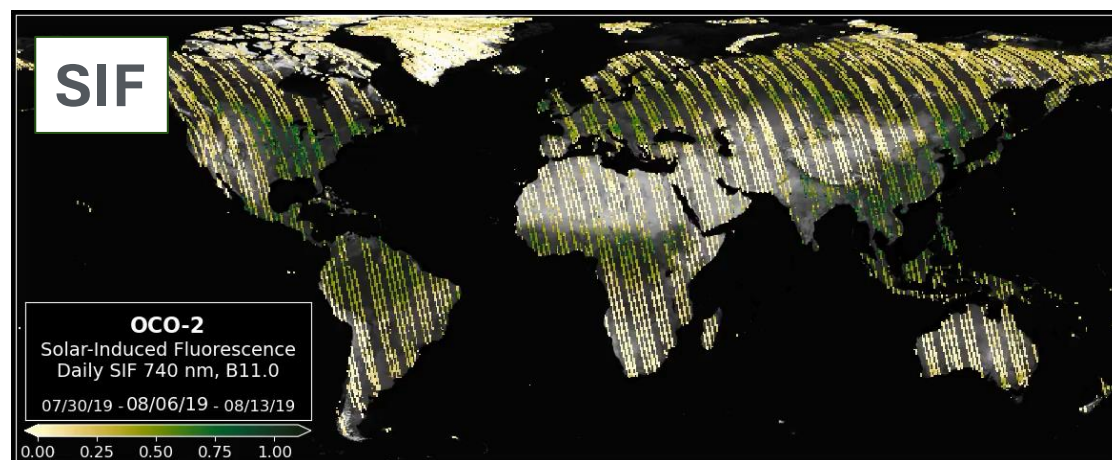
- Gold standard in spaceborne CO₂
- New insights into **natural and anthropogenic** components of carbon cycle
- **Global**, **regional**, and **local** scales

Solar-Induced chlorophyll Fluorescence (SIF)

- Proxy for photosynthesis
- Instantaneous view of plant health/stress
- Complementary to vegetation indices (e.g., NDVI)

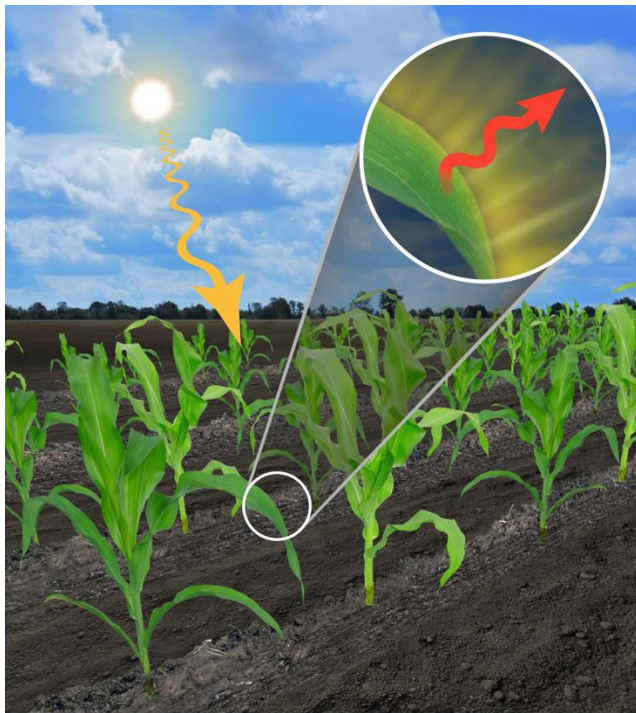
OCO-2 products serve as benchmarks for other current and future missions in the program of record.

- **XCO₂**: OCO-3, GOSAT, GOSAT-2, GOSAT-GW, CO2M
- **SIF**: OCO-3, TROPOMI, GOME-2, TEMPO



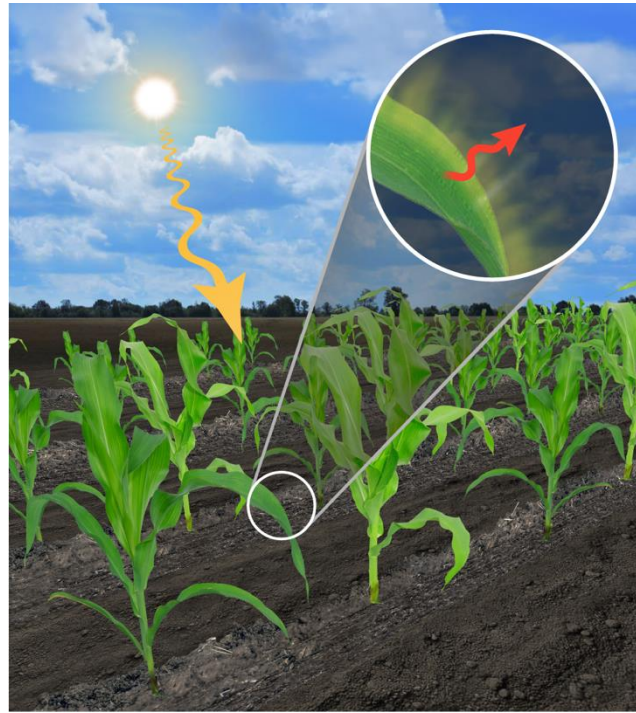
OCO-2 SIF is emerging as a critical indicator of ecosystem health.

Healthy



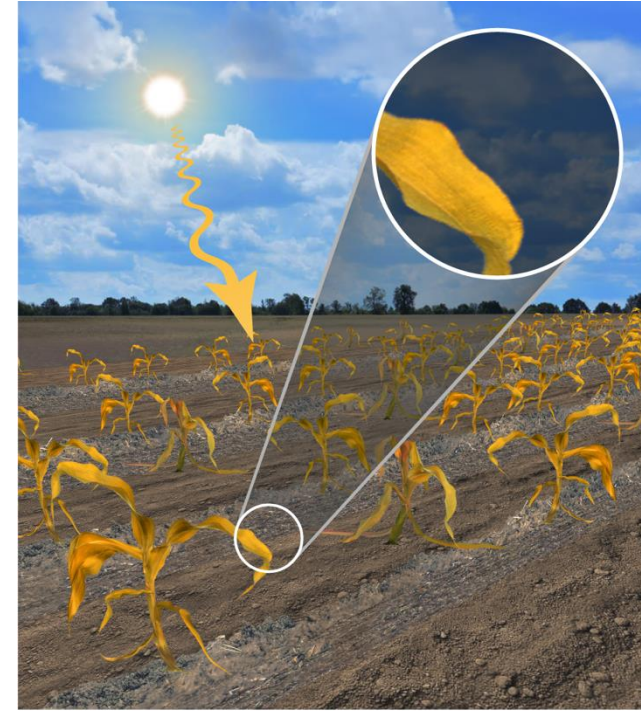
High SIF
High Greenness
No Stress

Appears Healthy



Reduced SIF
High Greenness
Increased Stress

Not Healthy/Mortality

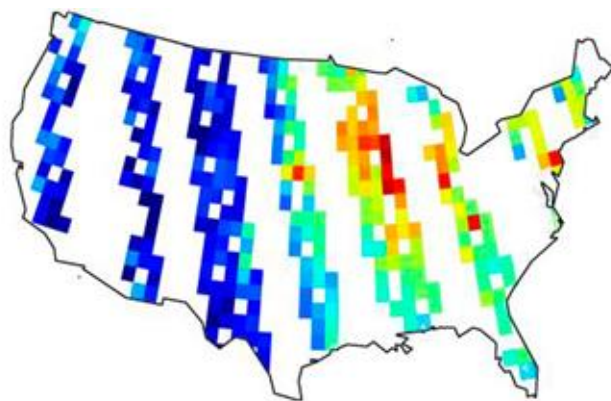


Low SIF
Low Greenness
High Stress



High-Resolution Mapping, Value-Added Products from SIF

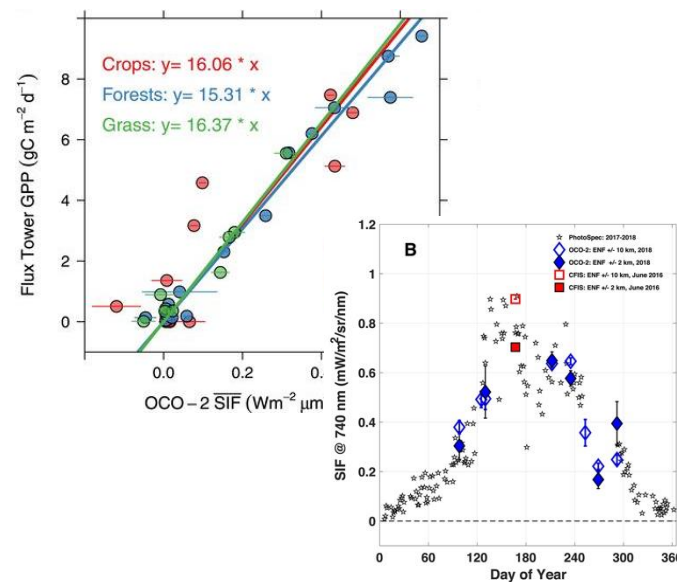
OCO-2 + OCO-3 SIF



OCO-2 Footprint: ~1 km
Size of a medium to large farm

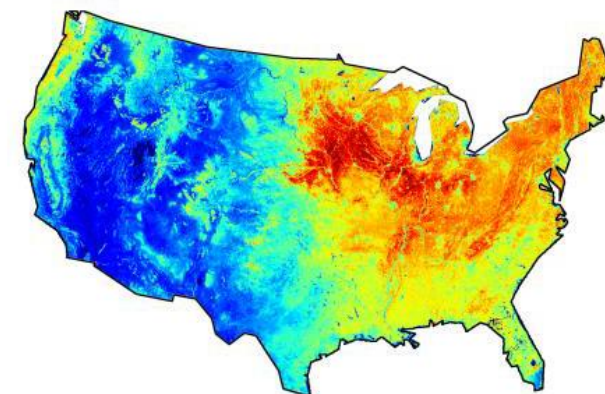
+

Traceability to Towers

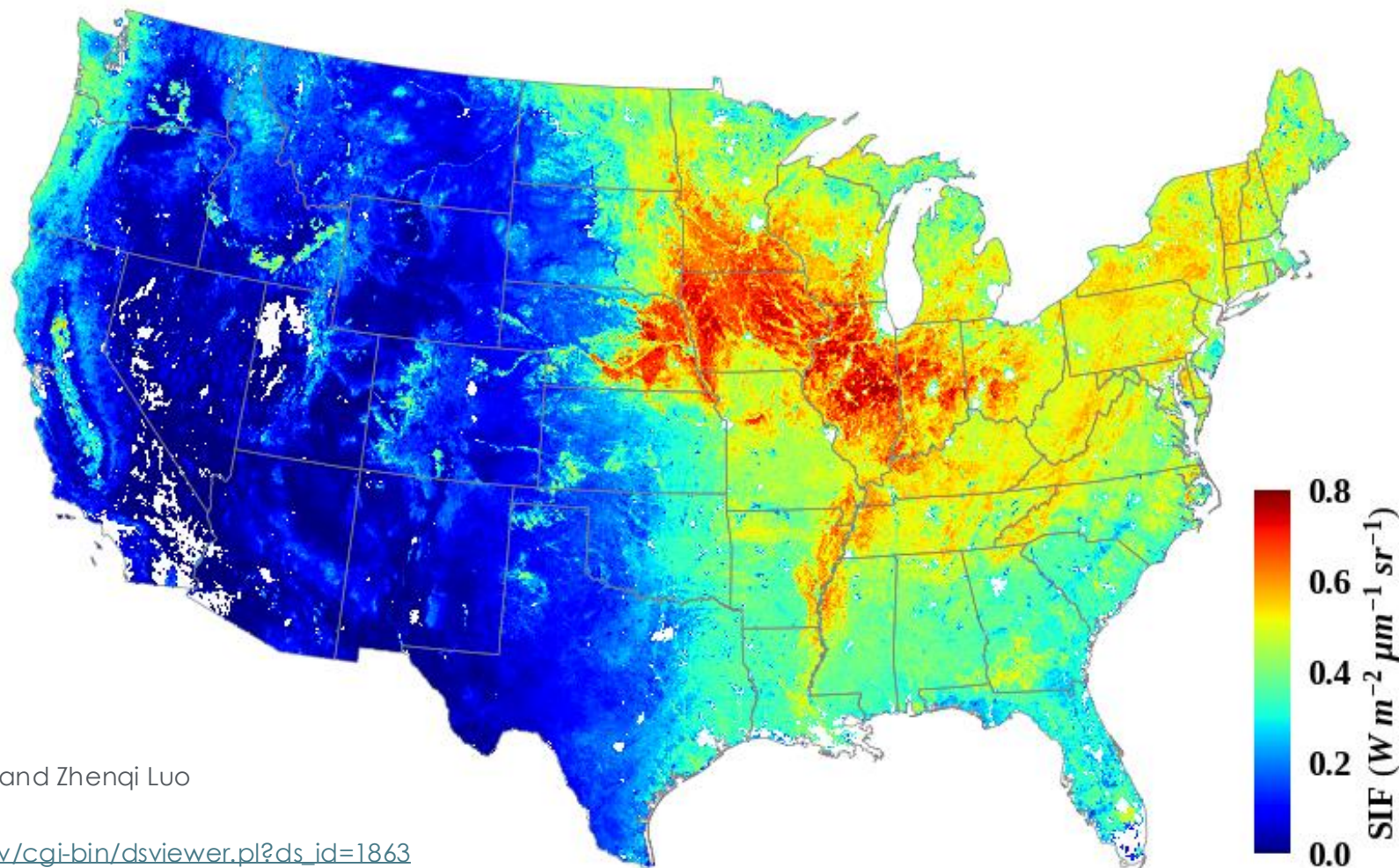


+

Ancillary Vegetation Data
(e.g., MODIS)



High-Resolution Mapping of Vegetation Activity



Courtesy Jiaming Wen and Zhenqi Luo

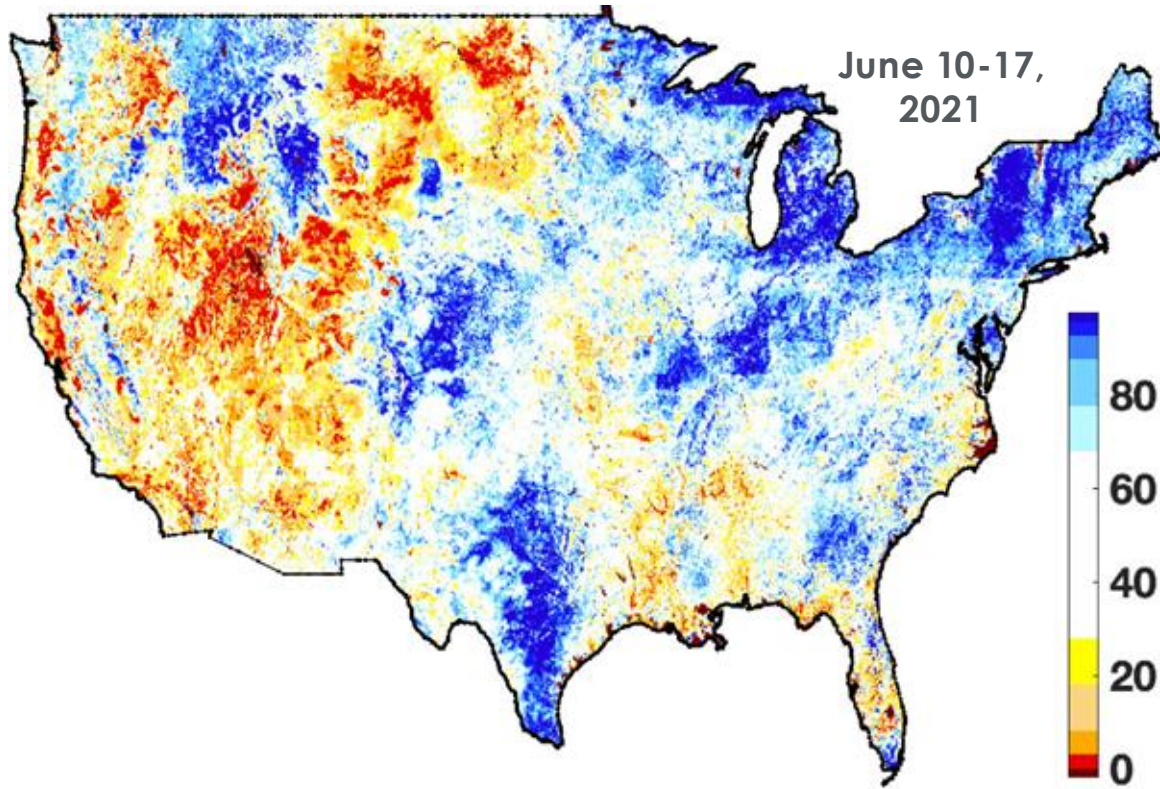
Data Availability:

- https://daac.ornl.gov/cgi-bin/dsviewer.pl?ds_id=1863

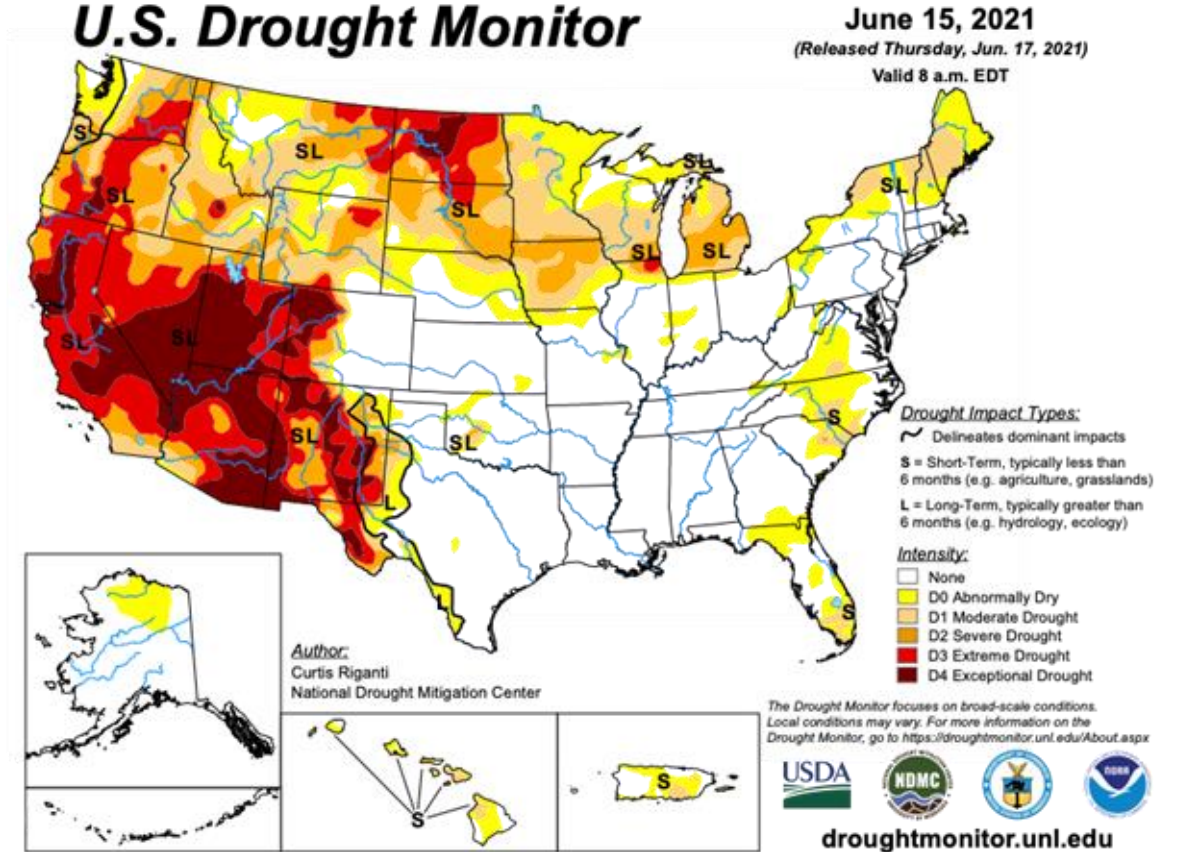


Drought Early Warning

SIF Percentile



U.S. Drought Monitor



OCO-2-based SIF products offer early warning information on flash drought of potential value to agencies such as the US Drought Monitor.



Together, OCO-2 and SMAP offer early warning and near-real-time tracking of flash drought.

A boost in plant productivity can foreshadow severe soil water loss.

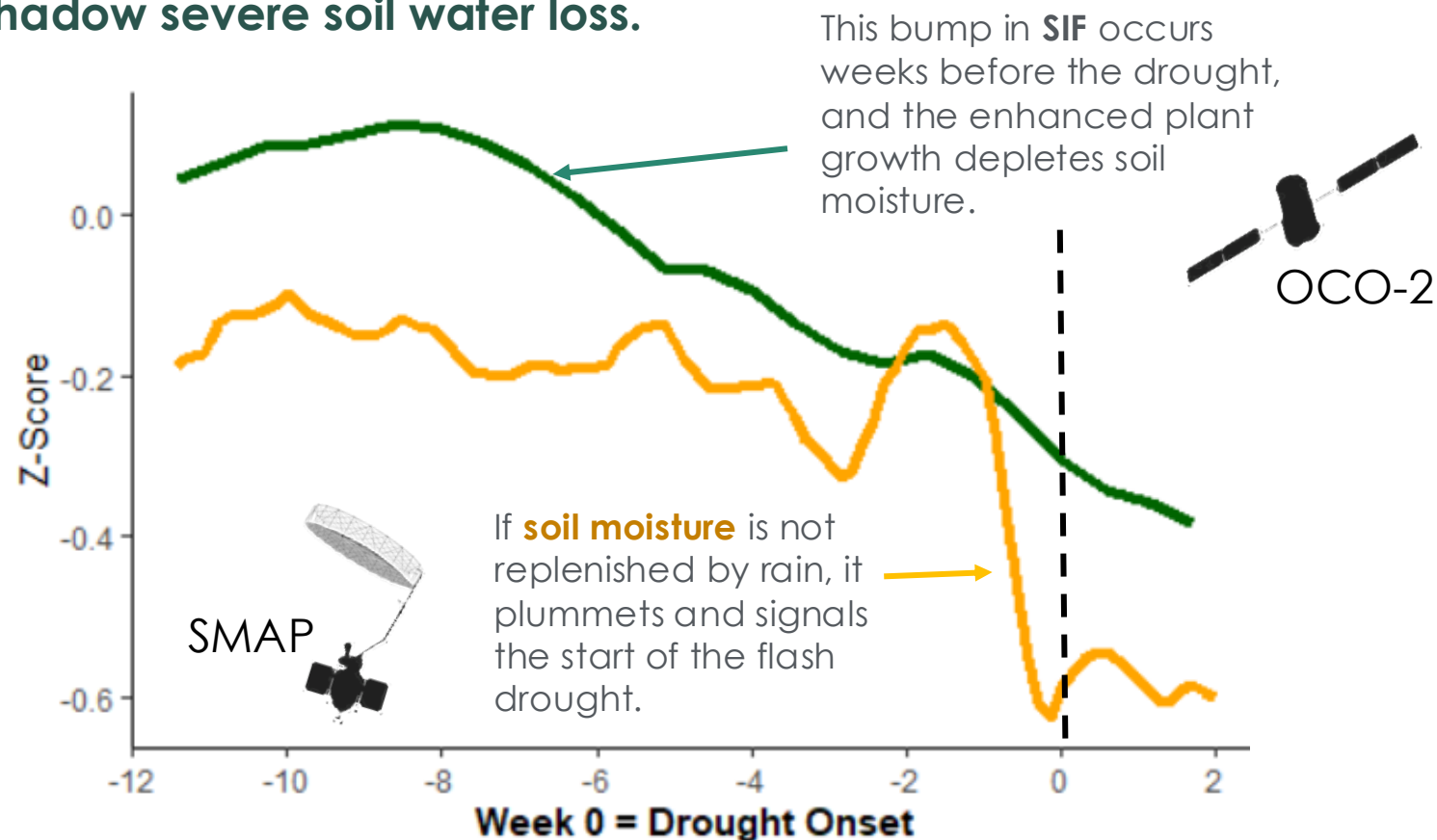
Flash Droughts

- Increase in frequency of extreme heat events has increased flash droughts in the US.
- By the time SMAP registers the dry soil, it's too late to prepare.

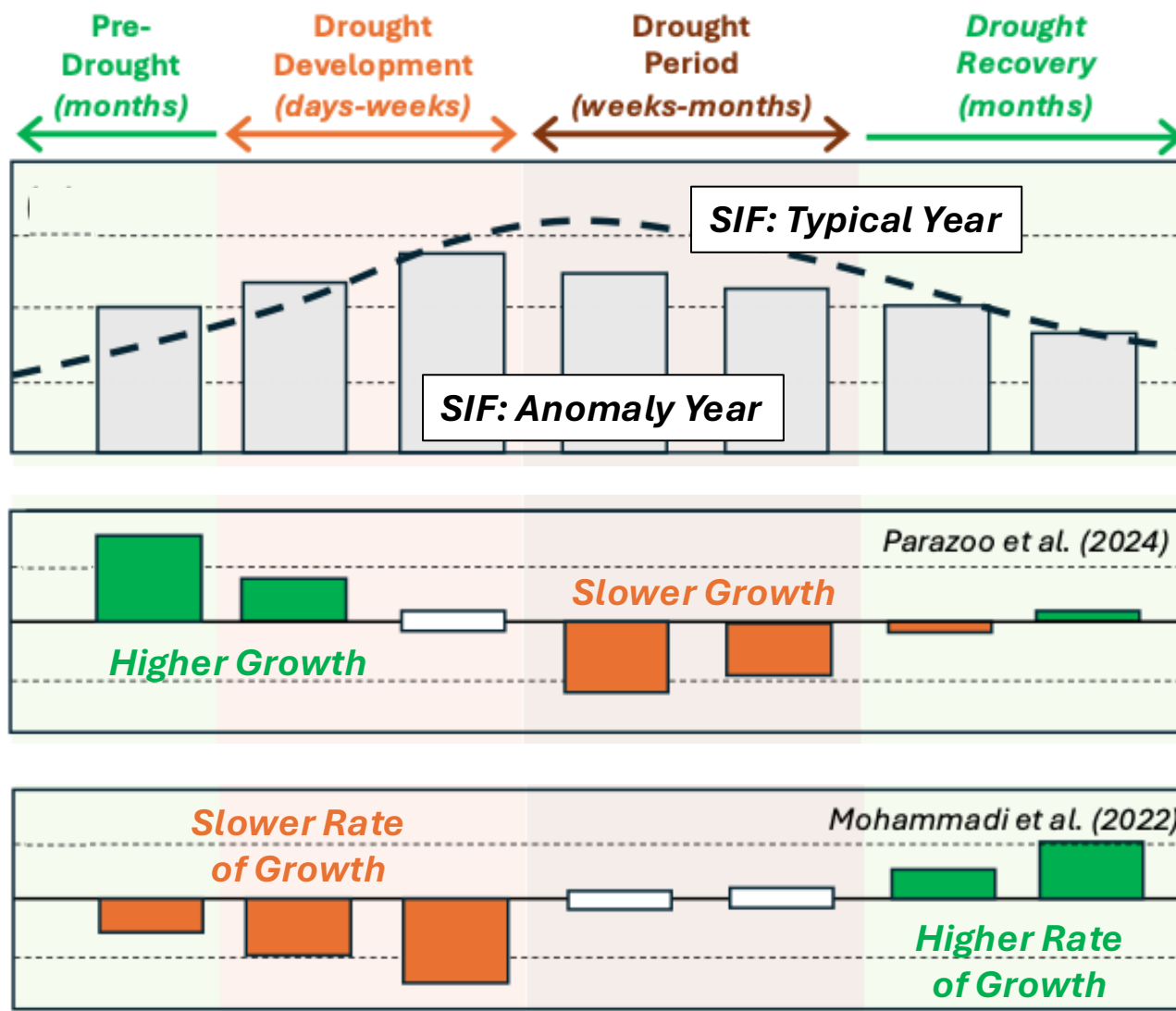
“Glowing” Early Warning

- OCO-2 SIF data show that early on, the warmth prompts a plant growth spurt. This enhances the “glow” from growing plants.
- This can signal a warning for flash drought up to 3 months ahead.

Parazoo et al., 2024



Introducing: SIF Drought Metrics



Idealized Drought

- SIF increases early
- Decline during drought development

Anomaly Index (Z-Score)

- Growth anomalies across drought

Rapid Change Index (SIF-RCI)

- Rate of change relative to preceding weeks and climatology



Summary of Which Drought Indices to Use

- **Short-Term Meteorological Drought:** SPI, PDSI
- **Persistent Meteorological Drought:** SPI, SPEI
- **Hydrological Drought:** SM
- **Agricultural, Ecological Drought:** SIF + SM
- **Flash Drought:** SIF + SM + Meteorological Indices (SPI, SPEI, PDSI)
- **Operational Decision Support:** Blended Indices (USDm)

The more information the better. Combining multiple drought indices and variables (meteorological, hydrological, ecological) offers the best chance for drought detection and attribution.



Tools and Data Hubs

Drought.gov (NIDIS) is a good starting place to find drought information.

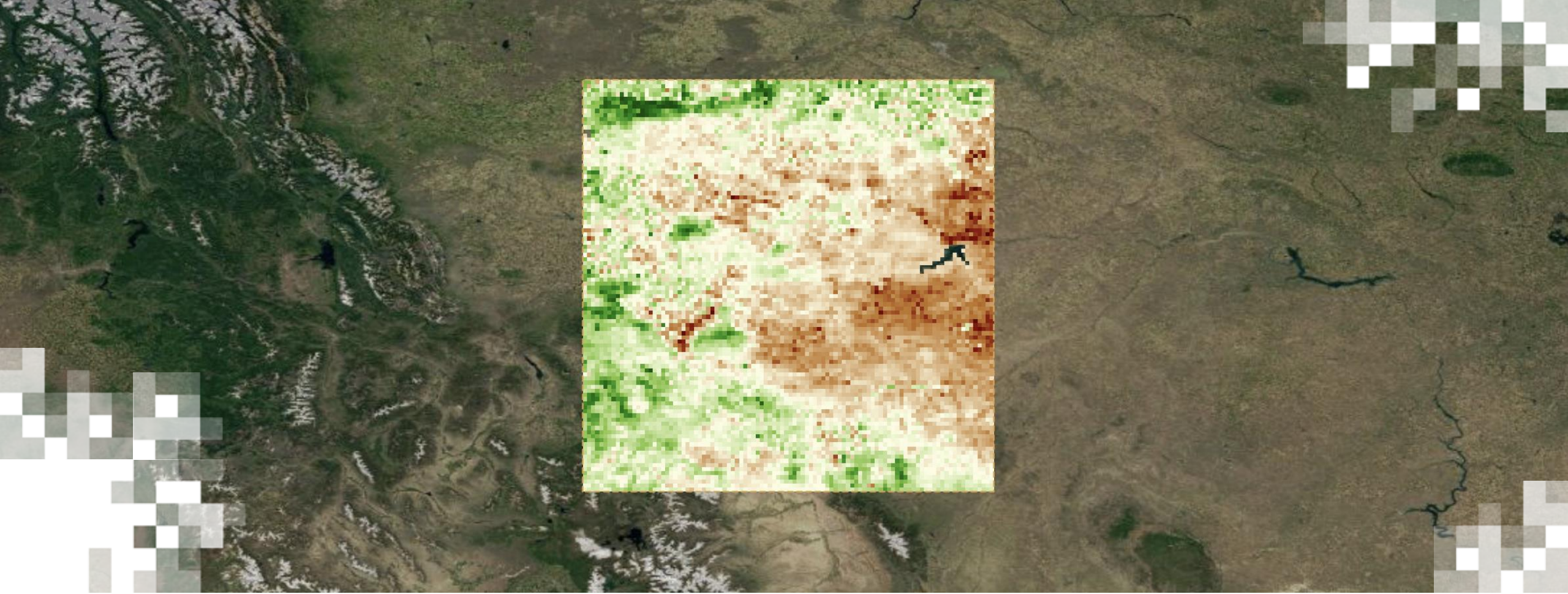
Standardized Drought Analysis Toolbox (Hao and AghaKouchak 2014; Farahmand and AghaKouchak 2015a): can be used to compute standardized drought indices, such as SSWI

Climate Engine (<https://app.climateengine.com/climateEngine>): Visualization of multiple datasets mentioned above and more, e.g., CHIRPS, CFS, CMORPH, ERA5, MERRA2, GLDAS, GPM, TRMM, NCEP reanalysis, OISST, gridMET, daymet, PRISM, etc. Satellite data such as MODIS, LandSat, and Sentinel are also available. Only selected variables are available depending on datasets.

The Climate Toolbox (<https://climatologytoolbox.org/>) at UC Merced

GMAO's MERRA-2 Visualization Tool (<https://fluid.nccs.nasa.gov/reanalysis/>)





Demo:
Flash Drought Indicators

Demo Outline

- **Session 1: Flash Drought Indicators**

- Solar-Induced Chlorophyll Fluorescence (SIF) and the SIF Rapid Change Index (SIF-RCI)
- Soil Moisture and the Soil Water Deficit Index (SWDI)

- **Session 2: Flash Drought Detection**

- Case Study: 2023/2024 Summer Flash Drought in South Africa
- Applying Detection Rules
- Investigating Spatial Distribution of Flash Drought

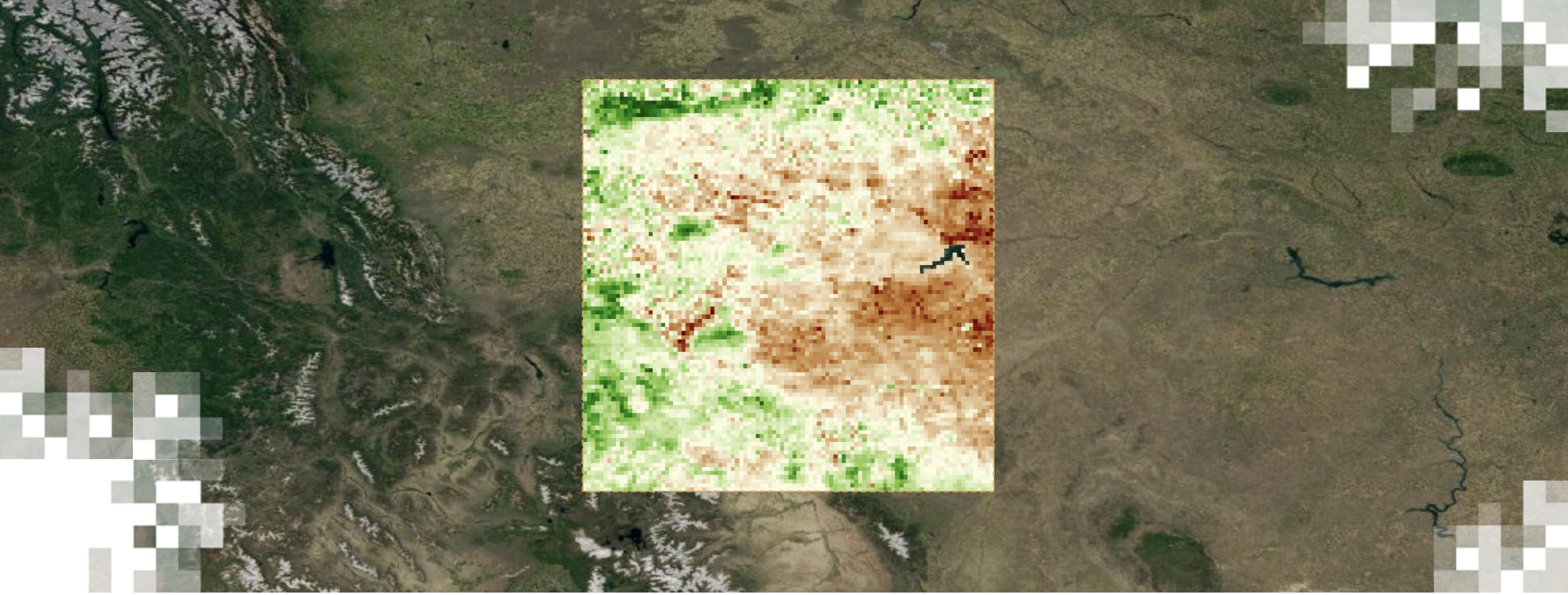


Session 1 Demo Objectives

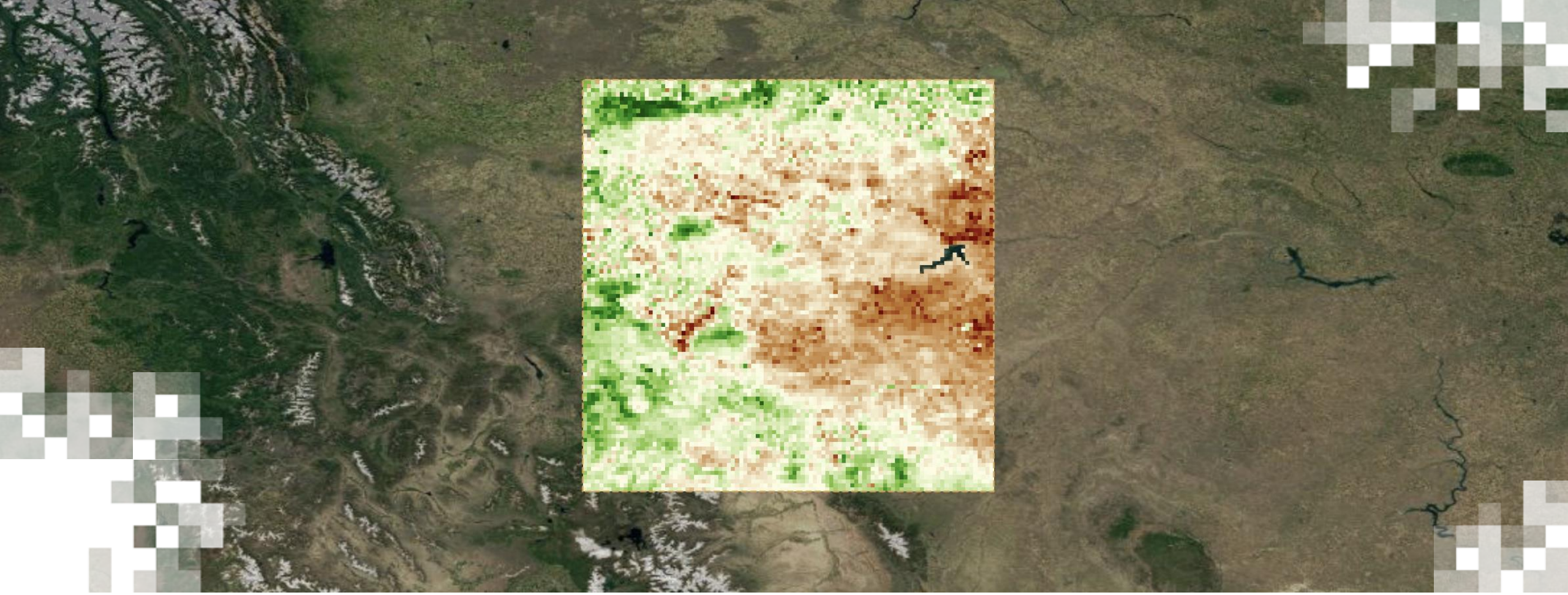
By the end of session 1, participants will be able to:

- Recognize how SIF data can be used as a drought indicator
- Retrieve a subset of GOSIF data and use it to derive a time series of SIF-RCI over a selected region
- Retrieve a subset of SMAP Level 4 Soil Moisture data and derive a time series of SWDI over a selected region
- Investigate how the SIF-RCI and SWDI time series can be used to study the development of the 2017 Northern Great Plains flash drought
- Compare the advantages and limitations of using SIF and Soil Moisture to study drought compared with meteorological data from related research





Getting Started with the Course Code



Session 1: Summary

Summary

- Flash drought is characterized by the rapid drying of land and unusually swift intensification of drought conditions. It often triggers or amplifies sustained drought and impacts ecosystems, crops, and human water management.
- SMAP provides daily mapping of near-surface soil moisture (top 5 cm) and model-integrated root-zone soil moisture (down to 100 cm).
- SMAP tracks the subsequent gradual and rapid draw-down of soil water leading into full flash drought development.
- OCO-2 measures SIF, an instantaneous indicator of plant photosynthetic activity, which acts as an early warning signature up to three months before flash drought onset by detecting plant stress and decreased photosynthetic activity before leaves turn brown or lose greenness.
- SIF anomalies are quantified using indices like the Z-score (standardized anomaly from the long-term mean) and the Rapid Change Index (growth rate change relative to preceding weeks, optimal for operational monitoring).
- No single index or process accounts for flash drought; combining multiple hydro-ecological indices (e.g., USDM, SMAP soil moisture, and OCO-2 SIF) provides the most reliable detection, attribution, and early warning capability.
- SIF Rapid Change Index (SIF RCI) and Soil Water Deficit Index (SWDI) can detect flash drought onset earlier than traditional monitoring tools like the US Drought Monitor.



Contact Information

Trainers:

- Nick Parazoo
nicholas.c.parazoo@jpl.nasa.gov
- Jackie Ryan
jacqueline.ryan@jpl.nasa.gov

- [ARSET Website](#)
- [ARSET YouTube](#)

For questions, comments, or to share how you have applied our trainings to your work or studies, email nasa.arset@gmail.com.

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Session 2

Flash Drought Detection and Prediction

- Synthesize Solar-Induced Fluorescence Rapid Change Index (SIF-RCI) time series data for a selected region using a provided Jupyter Notebook.
- Retrieve and manipulate SMAP Soil Moisture (SM) and land process model-based Soil Water Deficit Index (SWDI) time series and compare with SIF-RCI time series data using a provided Jupyter Notebook.
- Compare the implications of different methods for flagging and identifying flash droughts in real-world scenarios.





Thank You!



Accessing Course Materials

<https://github.com/jackiryan/flash-drought-ARSET>

